

Controlled Class-Imbalance Construction for TCGA-UT

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Abstract

Class imbalance in computational pathology is not only a question of how many examples a class contributes. A class may perform poorly because it is rare, because its frozen foundation-model features overlap with other classes, or because both effects coincide. The companion TCGA-UT imbalance benchmark established the dataset, the mitigation taxonomy, and patch-level and WSI-bag evaluations under the native TCGA-UT long tail. This report defines a controlled counterpart: artificial full-scale TCGA-UT training distributions that keep all 32 cancer types, preserve fixed validation and test splits, and vary only the training support assigned to each class. Power-law training distributions are generated under native-prevalence, easy-to-difficult, and difficult-to-easy class orders, using frozen Virchow2 patch features and WSI-bag features with the same 30-instance evidence budget as the companion benchmark. The design separates imbalance severity from the identity of the classes assigned to the head and tail of the training distribution. Discrimination and calibration are evaluated for patch-feature and WSI-bag mitigation methods in their native regimes, including post-hoc temperature scaling fitted on validation predictions.

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1 Introduction

The companion TCGA-UT imbalance benchmark compared class-imbalance mitigation methods in their native input regimes: labeled patch-feature classification and weakly supervised WSI-bag classification. It also showed that class frequency alone is an incomplete explanation for poor classwise performance. Some low-support cancer types remain separable under a frozen pathology foundation model, whereas other classes are difficult even when their support is not the smallest. This report turns that observation into an experimental design question: how should mitigation methods be interpreted when class frequency and inherent class difficulty can be varied deliberately?

TCGA-UT is a useful test bed for this question because it provides both slide labels and patch-level evidence derived from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). The native dataset already has a long-tailed slide distribution over 32 cancer types. Native imbalance is realistic, but it entangles prevalence with many other factors: tumor morphology, tissue diversity, slide acquisition, and the representation geometry induced by the frozen encoder. A controlled construction complements the native benchmark by imposing known training-support profiles while holding validation and test data fixed.

The goal is not to introduce a new mitigation method. Instead, the report evaluates representative methods from the companion taxonomy under constructed training distributions. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, and Scholz-style metric losses (Scholz et al., 2024). WSI-bag methods include MIL cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The construction starts from the same TCGA-UT case-disjoint split protocol as the companion benchmark. For each seed, the validation and test splits are kept fixed, and artificial imbalance is applied only to the training split. The split assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one of train, validation, or test. This isolates the effect of training support: model selection and held-out evaluation remain comparable across imbalance parameters and mitigation methods without patient leakage.

2.1 Full-Scale Training Pool

All 32 TCGA-UT cancer types are retained. Let $C = 32$ and let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool. For class c , let a_c be the number of available training slides before artificial resampling. The constructed training split must satisfy three constraints:

$$1 \leq n_c \leq a_c, \quad \sum_{c=1}^C n_c = |\mathcal{D}_{\text{train}}|, \quad c \in \{1, \dots, C\}. \quad (1)$$

The first constraint keeps every class represented and prevents oversampling beyond available slides. The second preserves the full training split size. The third makes the benchmark a 32-class TCGA-UT study throughout.

After slide selection, the input evidence is capped in the same way as in the companion benchmark. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024). Features are read from the shared `cls_patchmean` Virchow2 store (`concat(CLS, mean(patch tokens))`), 2,560-dimensional

vectors) used by the companion WSI-bag benchmark. The cap prevents slides with more extracted tumor patches from dominating the training objective while keeping the input budget aligned across regimes.

2.2 Power-Law Training Support

For an ordered class list $(c_1, \dots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^C j^{-\lambda}}. \quad (2)$$

When $\lambda = 0$, the intended distribution is uniform over ranks. As λ increases, early-ranked classes receive more training support and late-ranked classes receive less. Raw targets $|\mathcal{D}_{\text{train}}|p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder.

The target distribution is then projected onto the feasible set in Equation (1). Each class receives one guaranteed slide. Remaining slides are allocated according to Equation (2); classes whose requested support exceeds their available training support are capped, and the unused mass is redistributed over the uncapped classes using the same power-law ranks. This cap-and-redistribute procedure keeps the constructed split full-scale without dropping any cancer type.

2.3 Class Order

The same imbalance severity can have different consequences depending on which classes become frequent. This report therefore uses three class-order regimes. In the *native-prevalence order*, classes are ranked by their TCGA-UT slide support, so the constructed head and tail follow the realistic direction of the native TCGA-UT long tail. In the *easy-to-difficult order*, classes are ranked by decreasing full-support baseline recall from the companion benchmark, averaged across the patch-feature CE and WSI-bag MIL-CE baselines. In the *difficult-to-easy order*, that ranking is reversed. The difficulty orders deliberately couple support with inherent class separability in opposite directions: the first assigns more support to classes that were already easier under balanced/full-support training, whereas the second assigns more support to classes that were difficult despite the frozen Virchow2 representation.

The construction therefore separates support magnitude from the held-out evaluation distribution while also varying the semantic identity of head and tail classes. Frequency is imposed by λ and rank; validation and test cases are unchanged. The companion report’s class-difficulty analysis remains the interpretive background: if a class performs poorly when it is assigned high training support, scarcity is not the only plausible explanation.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or feature-space bag augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, or prototype-affinity objectives. WSI-specific methods change how slide bags are represented or mixed.

Let y be the class label, $\mathbf{z} \in \mathbb{R}^C$ the logits, and p_y the softmax probability of the true class. Cross-entropy optimizes $-\log p_y$. Weighted cross-entropy multiplies this term by a normalized inverse-frequency weight. Focal loss uses

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^\gamma \log p_y, \quad (3)$$

Regime	Family	Methods	Controlled question
Patch	Baseline and losses	CE, weighted CE, focal loss	Frequency weighting and hard-example emphasis.
Patch	Metric losses	CE + soft F1, CE + soft MCC	Macro-metric losses with balanced minibatch exposure.
Patch	Specialized	CFAL, divide-and-conquer	Prototype-affinity and expert fusion objectives.
WSI bag	Baseline and losses	MIL CE, weighted CE, focal loss	Slide-level analogues of standard losses.
WSI bag	MIL-specific	Balanced sampling, RankMix, SC-MIL, MDE-MIL	Bag-level sampling, augmentation, contrastive structure, and expert training.

Table 1: Method groups evaluated under constructed TCGA-UT training distributions. Patch and WSI-bag methods are compared only within their own input regime.

so confidently classified examples contribute less to the gradient. The soft-F1 and soft-MCC variants add differentiable approximations of macro evaluation metrics to CE, following the imbalance-aware medical-image objectives of Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are retained as WSI-bag methods because their mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

4 Evaluation Protocol

Each constructed split is indexed by seed, class-order regime, and imbalance parameter λ . The main reporting points are $\lambda = 0.0$, $\lambda = 1.0$, and $\lambda = 1.3$. For every method and every constructed regime (class order $\times \lambda$), hyperparameters are selected by validation macro F1 across three seeds; ties are broken by validation balanced accuracy. Baseline CE and MIL CE use fixed protocol settings. Selected configurations are evaluated on the fixed test split. Macro F1 is the primary discrimination metric because it weights cancer types equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics. Calibration is summarized with negative log-likelihood, multiclass Brier score, and expected calibration error. For each trained model, a scalar temperature is fitted on the validation probabilities and applied to the corresponding test probabilities, so raw and temperature-scaled calibration can be compared without changing discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

5 Results

The result analysis asks how methods behave as the native-prevalence support profile is strengthened. Table 2 and Figure 1 show the imposed support profile. Tables 3 and 4 then compare mitigation methods within the patch-feature and WSI-bag regimes.

At the patch-feature level, the constructed imbalance mainly affects balanced accuracy rather than overall accuracy. Plain CE remains accurate under all three settings, but its balanced accuracy falls from 0.857 at $\lambda = 0.0$ to 0.814 at $\lambda = 1.3$, and macro F1 falls from 0.834 to 0.819. Focal loss follows nearly the same pattern, indicating that hard-example emphasis alone does not correct the constructed support skew in this frozen-feature setting. In contrast, class weighting, balanced sampling, and CE combined with soft-F1 or soft-MCC keep

Order	λ	Training slides	Support range
native_prevalence	0.0	6110	190.3–191.3
native_prevalence	1.0	6110	48.0–1498.7
native_prevalence	1.3	6110	25.0–2204.0

Table 2: Constructed training-support summaries by class-order regime and imbalance severity.

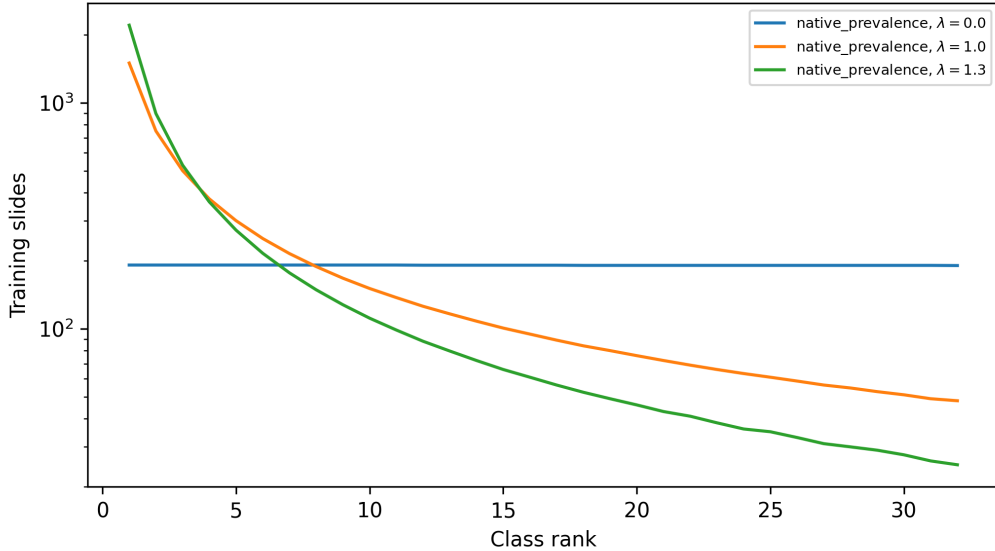


Figure 1: Constructed training support by class rank for the main imbalance settings.

balanced accuracy near 0.854–0.855 at $\lambda = 1.3$. The metric-loss variants give the highest patch macro F1 at the strongest reported imbalance (0.836), but the differences between the successful frequency-aware patch methods are small.

The WSI-bag regime shows a stronger separation between method families. MIL CE declines with increasing imbalance, from 0.823 macro F1 at $\lambda = 0.0$ to 0.802 at $\lambda = 1.3$. Weighted MIL improves over CE at $\lambda = 0.0$ and $\lambda = 1.0$, but it also declines at $\lambda = 1.3$. Balanced sampling and focal MIL are comparatively stable at the strongest imbalance, but neither dominates the baseline across all severities. The strongest WSI-bag method is MDE-MIL: it reaches the highest macro F1 at every reported λ (0.859, 0.858, and 0.842) and also the highest balanced accuracy at $\lambda = 0.0$. RankMix is the closest competitor, with balanced accuracy 0.866 at $\lambda = 0.0$ and 0.844 at $\lambda = 1.3$. These results mirror the companion benchmark’s conclusion that WSI-specific mechanisms can matter beyond simple class reweighting, because the bag-level task includes instance aggregation as well as class-frequency imbalance.

6 Discussion

The controlled construction gives frequency a known experimental role. Under native-prevalence ordering, increasing λ strengthens the same head-to-tail direction present in TCGA-UT while preserving the validation and test distributions. The CE baselines in both regimes lose balanced performance as λ increases, showing that the constructed support profile is strong enough to change class-balanced discrimination even though the frozen Virchow2 features remain informative.

Patch-feature mitigation behaves mostly like a correction of class exposure or loss contribution. At $\lambda = 1.3$, class weighting, balanced sampling, and the two CE-plus-metric objectives

Method	Accuracy	Balanced accuracy	Macro F1
balanced_sampler (native_prevalence, $\lambda = 0.0$)	0.875 ± 0.010	0.857 ± 0.010	0.832 ± 0.011
balanced_sampler (native_prevalence, $\lambda = 1.0$)	0.873 ± 0.016	0.852 ± 0.011	0.830 ± 0.016
balanced_sampler (native_prevalence, $\lambda = 1.3$)	0.875 ± 0.008	0.855 ± 0.006	0.834 ± 0.004
ce (native_prevalence, $\lambda = 0.0$)	0.877 ± 0.012	0.857 ± 0.010	0.834 ± 0.015
ce (native_prevalence, $\lambda = 1.0$)	0.879 ± 0.007	0.834 ± 0.002	0.828 ± 0.006
ce (native_prevalence, $\lambda = 1.3$)	0.871 ± 0.010	0.814 ± 0.004	0.819 ± 0.005
ce_soft_f1 (native_prevalence, $\lambda = 0.0$)	0.879 ± 0.010	0.860 ± 0.011	0.838 ± 0.013
ce_soft_f1 (native_prevalence, $\lambda = 1.0$)	0.876 ± 0.014	0.853 ± 0.009	0.833 ± 0.015
ce_soft_f1 (native_prevalence, $\lambda = 1.3$)	0.877 ± 0.007	0.854 ± 0.006	0.836 ± 0.003
ce_soft_mcc (native_prevalence, $\lambda = 0.0$)	0.879 ± 0.009	0.860 ± 0.010	0.837 ± 0.012
ce_soft_mcc (native_prevalence, $\lambda = 1.0$)	0.877 ± 0.013	0.854 ± 0.008	0.834 ± 0.013
ce_soft_mcc (native_prevalence, $\lambda = 1.3$)	0.877 ± 0.007	0.854 ± 0.006	0.836 ± 0.003
focal (native_prevalence, $\lambda = 0.0$)	0.875 ± 0.012	0.852 ± 0.009	0.834 ± 0.013
focal (native_prevalence, $\lambda = 1.0$)	0.877 ± 0.007	0.832 ± 0.001	0.829 ± 0.002
focal (native_prevalence, $\lambda = 1.3$)	0.870 ± 0.010	0.813 ± 0.006	0.819 ± 0.005
weighted_ce (native_prevalence, $\lambda = 0.0$)	0.877 ± 0.012	0.857 ± 0.010	0.834 ± 0.015
weighted_ce (native_prevalence, $\lambda = 1.0$)	0.877 ± 0.009	0.855 ± 0.008	0.832 ± 0.009
weighted_ce (native_prevalence, $\lambda = 1.3$)	0.874 ± 0.010	0.854 ± 0.004	0.832 ± 0.003

Table 3: Patch-feature test results for validation-selected configurations under constructed training distributions.

recover roughly four balanced-accuracy points relative to CE, while focal loss remains close to CE. This suggests that the main patch-level failure mode under constructed prevalence skew is not simply an abundance of easy examples, but the unequal contribution of class supports to the optimization objective. The successful patch methods do not create large separations from each other, which is consistent with the companion report’s observation that a strong frozen encoder compresses many patch-level method differences.

WSI-bag mitigation does not reduce to the same pattern. MDE-MIL and RankMix remain ahead of plain MIL CE under constructed imbalance, whereas balanced sampling and focal MIL provide smaller and less uniform gains. This points to an aggregation-specific component of the imbalance problem: slide-level prediction depends on which feature instances are emphasized inside each bag, not only on how often each class appears in minibatches. MDE-MIL’s advantage across all reported λ values suggests that coupling distribution-specific experts is useful when the training distribution is deliberately skewed, while RankMix remains competitive by augmenting informative bag subsequences.

These findings should still be interpreted within the native-prevalence construction reported here. The companion benchmark showed that class frequency is not the whole story; the present experiment shows that even when frequency is controlled directly, the best mitigation mechanism depends on prediction unit. Patch classifiers benefit from straightforward frequency-aware objectives, whereas WSI-bag classifiers benefit more from methods that alter bag representation or bag-level training dynamics.

7 Conclusion

This report defines a full-scale controlled TCGA-UT imbalance benchmark that keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies only the training support profile. Under native-prevalence ordering, increasing imbalance lowers the balanced performance of CE baselines in both patch-feature and WSI-bag regimes. Patch-level class weighting, balanced sampling, and CE plus soft-F1 or soft-MCC largely recover this loss, while focal loss does not. At slide level, MDE-MIL and RankMix are the strongest methods, indi-

Method	Accuracy	Balanced accuracy	Macro F1
mde_mil (native_prevalence, $\lambda = 0.0$)	0.890 ± 0.002	0.867 ± 0.011	0.859 ± 0.012
mde_mil (native_prevalence, $\lambda = 1.0$)	0.886 ± 0.009	0.851 ± 0.015	0.858 ± 0.012
mde_mil (native_prevalence, $\lambda = 1.3$)	0.879 ± 0.010	0.836 ± 0.019	0.842 ± 0.015
mil_balanced_sampler_ce (native_prevalence, $\lambda = 0.0$)	0.860 ± 0.010	0.825 ± 0.005	0.821 ± 0.005
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.0$)	0.855 ± 0.016	0.807 ± 0.023	0.808 ± 0.016
mil_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.865 ± 0.006	0.829 ± 0.020	0.828 ± 0.016
mil_ce (native_prevalence, $\lambda = 0.0$)	0.862 ± 0.001	0.834 ± 0.020	0.823 ± 0.020
mil_ce (native_prevalence, $\lambda = 1.0$)	0.857 ± 0.013	0.816 ± 0.019	0.813 ± 0.021
mil_ce (native_prevalence, $\lambda = 1.3$)	0.855 ± 0.020	0.812 ± 0.037	0.802 ± 0.037
mil_focal (native_prevalence, $\lambda = 0.0$)	0.848 ± 0.014	0.818 ± 0.003	0.807 ± 0.006
mil_focal (native_prevalence, $\lambda = 1.0$)	0.856 ± 0.015	0.809 ± 0.008	0.809 ± 0.007
mil_focal (native_prevalence, $\lambda = 1.3$)	0.858 ± 0.010	0.811 ± 0.007	0.817 ± 0.002
mil_weighted_ce (native_prevalence, $\lambda = 0.0$)	0.863 ± 0.005	0.842 ± 0.010	0.828 ± 0.015
mil_weighted_ce (native_prevalence, $\lambda = 1.0$)	0.869 ± 0.013	0.836 ± 0.017	0.832 ± 0.015
mil_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.854 ± 0.009	0.814 ± 0.002	0.810 ± 0.009
rankmix_mil (native_prevalence, $\lambda = 0.0$)	0.882 ± 0.003	0.866 ± 0.008	0.841 ± 0.008
rankmix_mil (native_prevalence, $\lambda = 1.0$)	0.875 ± 0.011	0.851 ± 0.020	0.835 ± 0.025
rankmix_mil (native_prevalence, $\lambda = 1.3$)	0.878 ± 0.009	0.844 ± 0.017	0.833 ± 0.016
sc_mil (native_prevalence, $\lambda = 0.0$)	0.854 ± 0.006	0.826 ± 0.008	0.811 ± 0.004
sc_mil (native_prevalence, $\lambda = 1.0$)	0.860 ± 0.019	0.822 ± 0.023	0.819 ± 0.022
sc_mil (native_prevalence, $\lambda = 1.3$)	0.859 ± 0.018	0.819 ± 0.018	0.822 ± 0.017

Table 4: WSI-bag test results for validation-selected configurations under constructed training distributions.

Method	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
Patch calibration table unavailable.							

Table 5: Patch-feature calibration before and after validation-fitted temperature scaling.

cating that WSI-bag imbalance mitigation must account for aggregation and bag composition rather than only class frequency. The resulting analyses connect directly to the companion native long-tail benchmark while asking a more controlled question: when mitigation helps, is it correcting scarcity, changing the aggregation problem, or redistributing performance across the class spectrum?

Method	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
WSI-bag calibration table unavailable.							

Table 6: WSI-bag calibration before and after validation-fitted temperature scaling.

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